

16.3 Frequency division multiple access

Analogue transmission is considered as an "older" cellular phone technology. Analogue allows a cellular phone to transmit signals by sending voice, video, and data that are always changing.

The FDMA is the simplest scheme used to provide multiple access in analogue transmission. In FDMA systems, the radio frequency spectrum is divided into several frequency bands separated by a certain guard band. Each frequency band can be used simultaneously.

In this technique, the bandwidth is divided into a number of channels and distributed among users with a finite portion of bandwidth for permanent use as illustrated in Figure 16.8. In addition, we can observe from Figure 16.8 that FDMA permits only one user per channel because it allows the user to use the channel 100 per cent of the time. Therefore, only the frequency

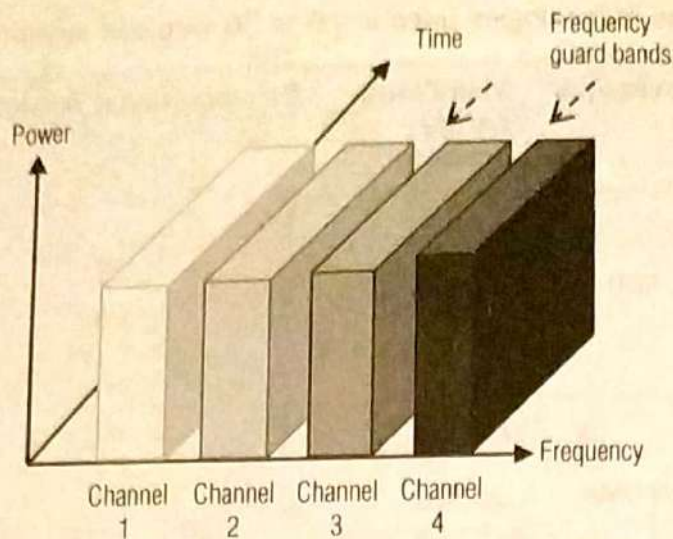


Figure 16.8 FDMA principle of operation

"dimension" is used to define channels. Each block represents a different user. Figure 16.9 shows how the three users occupy the time and the bandwidth with FDMA.

Frequency guard bands are provided between adjacent signal spectra to minimize crosstalk between adjacent channels.

In FDMA, the channel has two frequencies, namely forward channel and reverse channel. When the FDMA technique is employed, as long as the user is engaged in "conversation," no other user can access the same spectrum space.

Number of channels supported by FDMA system

The number of channels that can be simultaneously supported in FDMA system is given by (Figure 16.10)

$$\text{Number of channels } (N) = \frac{B_t - 2B_g}{B_c} \quad (16.1)$$

where

B_t is the allocated frequency spectrum

B_g is the guard band allocated at the edge of frequency spectrum

B_c is the channel bandwidth

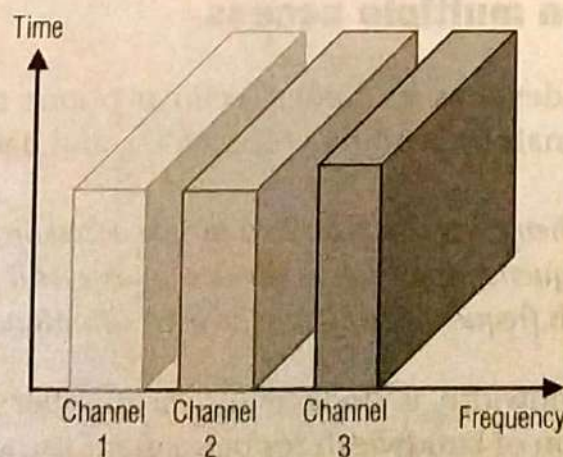


Figure 16.9 Time and bandwidth occupancy of three user signals with FDMA

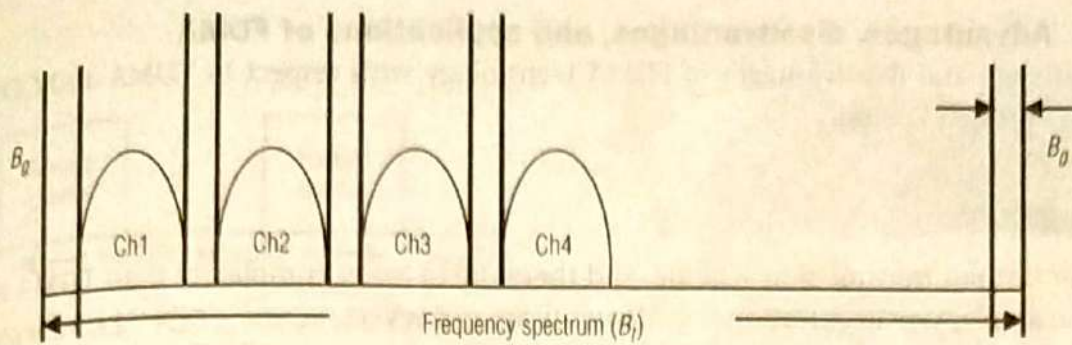


Figure 16.10 Frequency spectrum of FDMA

Example problem 16.1

If each cellular carrier is allocated N number of channels and the allocated frequency spectrum is 12.5 MHz, guard band is 10 KHz and the bandwidth of each channel is 30 KHz. Find the total number of channels provided by the cellular carrier.

Solution

From Equation (16.1)

$$\text{Number of channels } (N) = \frac{B_t - 2B_g}{B_c}$$

$$\text{Number of channels } (N) = \frac{[(12.5 \times 10^6) - 2(10 \times 10^3)]}{30 \times 10^3} = \mathbf{416}$$

Example problem 16.2

In the United States, the advanced mobile phone service (AMPS) cellular operator is allocated 15 MHz for each simplex band, and if B_t is 10 MHz, B_g is 10 KHz, and B_c is 25 KHz. Find the number of channels available in an FDMA system.

Solution

The number of channels available in the FDMA system is given as

$$N = \frac{B_t - 2B_g}{B_c}$$

$$N = \frac{10 \times 10^6 - 2(10 \times 10^3)}{25 \times 10^3} \approx 399$$

In the United States, each cellular carrier is allocated **399** channels.

16.3.1 Advantages, disadvantages, and applications of FDMA

The advantages and disadvantages of FDMA technology with respect to TDMA and CDMA are discussed in the following.

Advantages

1. A continuous transmission scheme, and therefore of lower complexity than TDMA scheme, for example, synchronization requirements are not severe
2. Simple to implement from a hardware standpoint, because multiple users are isolated by employing simple band pass filters
3. Fairly efficient with a small base population and when traffic is constant
4. No channel equalization required
5. Capacity can be increased by reducing the information bit rate and using an efficient digital speech coding scheme

Disadvantages

1. If an FDMA channel is not in use, then it sits idle and cannot be used by other users to increase or share the capacity. Therefore, FDMA implementation becomes *inefficient use of spectrum*.
2. FDMA requires tight RF filtering to minimize adjacent channel interference.
3. Network and spectrum planning are intensive.
4. Frequency planning is time consuming.
5. Even though no two users use the same frequency band at the same time, guard bands are introduced between frequency bands to minimize adjacent channel interference. Guard bands are unused frequency slots that separate neighbouring channels. This leads to a waste of bandwidth. When continuous transmission is not required, bandwidth goes wasted since it is not being utilized for a portion of the time.

Applications of FDMA

1. Walkie-talkies and mobile networks for closed user groups often use FDMA.
2. Another example of FDMA is AM or FM radio broadcasting, where each station has its own channel.
3. Early cellular telephony mostly used FDMA analogue transmission.

16.3.2 FDMA versus frequency division duplex

The pairing of channels is to provide a two-way communication in a direction, uplink or downlink (for instance, the traffic going back and forth between a mobile phone and a base-station), is termed as frequency division duplex (FDD) (Figure 16.11). The composite scheme, with separate frequency channels assigned for each direction of communication, is thus labelled FDMA/FDD (Figure 16.12).

FDD provides two simplex channels at the same time.

FDD is a physical layer technique that combines and transmits low-bandwidth channels through a high-bandwidth channel. FDMA, on the other hand, is an access method in the data link layer.

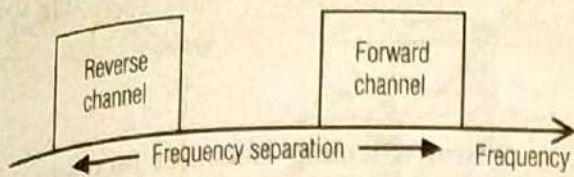


Figure 16.11 FDD provides two simplex channels

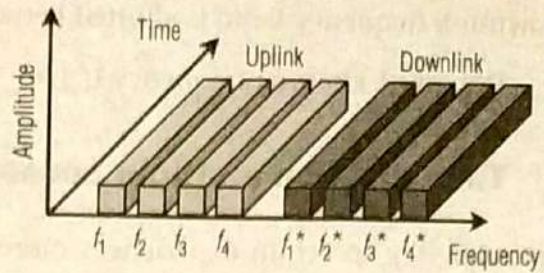


Figure 16.12 FDMA/FDD

16.3.3 Example and applications of FDMA in 1G cellular (or AMPS) and GSM 900 Applications of FDMA in 1G

1G cellular systems are analogue and they operate on the FDMA principle. The AMPS is 1G analogue cellular system used in North America and is an example of a pure FDMA. The AMPS system was allocated a frequency range of **869–894 MHz** for downlink or forward transmission from base station to mobile station and **824–849 MHz** band for the uplink or reverse communication from mobile station to base station. One channel in each direction, are assigned to one mobile user in these 25 MHz bands (894 MHz–869 MHz and 849 MHz–824 MHz). The assignment of frequency channels is sometimes referred to as a frequency-orthogonal assignment, since it enables multiple users to simultaneously use the system. Each user is accommodated a unique channel in AMPS. When a call is completed, or when a handoff occurs, the channel is vacated so that another mobile subscriber may use it.

Application of FDD/FDMA scheme in GSM 900 systems

In GSM 900, the frequencies for uplink and downlink communication are allotted by the base station as illustrated in Figure 16.13. The uplink frequency band is allotted between 890.2 and 915 MHz. The uplink channel frequency (f_u) allocation is done using

$$f_u = 890 \text{ MHz} + n \times 0.2 \text{ MHz} \quad (16.2)$$

where n is the channel number (starting from 1 to 124).

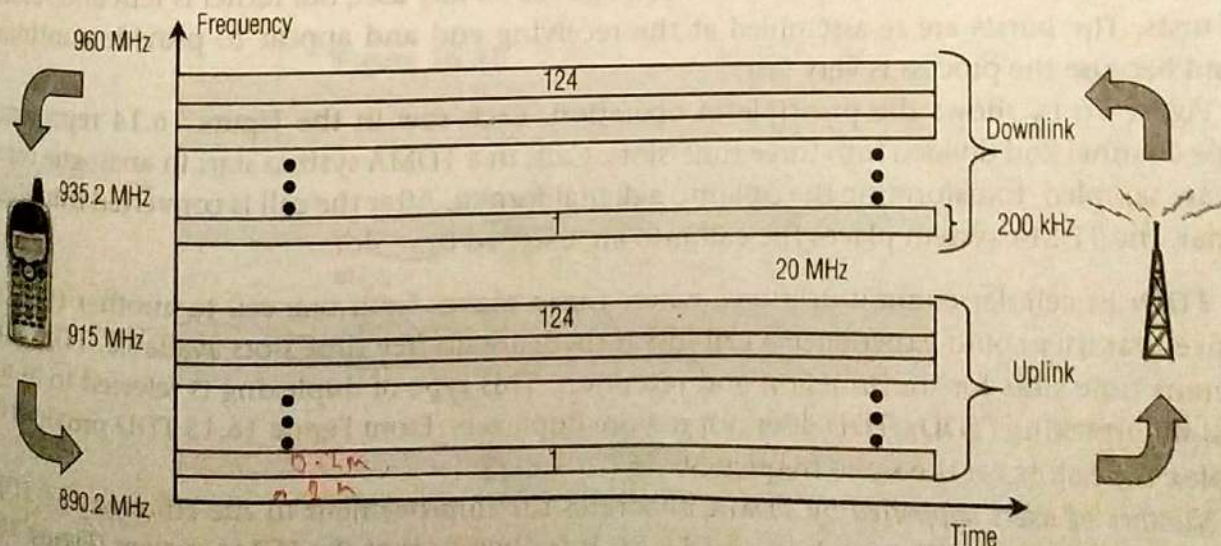


Figure 16.13 Illustration of frequencies for uplink and downlink communication

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The downlink frequency band is allotted between 935 and 960 MHz and is allocated using

$$\text{Downlink channel frequency } (f_d) = f_u + 45 \text{ MHz} = 935 \text{ MHz} + n \times 0.2 \text{ MHz} \quad (16.3)$$

16.4 Time division multiple access

As the frequency spectrum experiences more traffic, spectrum efficiency becomes increasingly important. TDMA systems were developed as FDMA system spectrum efficiency became insufficient. In digital systems, continuous transmission is not required because users do not use the allotted bandwidth all the time. It allows several users to share the same frequency band by dividing the timescale into different time slots which are periodically allocated to each mobile user for the duration of a call.

TDMA systems have the capability to split users into time slots because they transfer digital data, instead of analogue data commonly used in legacy FDMA systems.

Digital 2G cellular systems that used the TDMA technology are GSM, IS-136, PDC, and DECT standard for portable phones

16.4.1 TDMA principle of operation

TDMA systems divide the radio spectrum into time slots and each user is allowed to either transmit or receive in each time slots (i.e., different users can use the same frequency in the same cell but at different times).

TDMA multiplexes three signals over a single channel. The current TDMA standard for cellular divides a single channel into six time slots, with each signal using two slots, providing a three to one gain in capacity over AMPS. Each caller occupies a cyclically repeating time slots.

In TDMA, when the caller depresses the push-to-talk (PTT) switch, a control channel registers the radio to the closest base station. During registration, the base station assigns the user an available pair of channels, one to transmit and the other to receive. However, unlike an FDMA system registration, a TDMA system registration also assigns an available time-slot within the channel. The user can only send or receive information at that time, regardless of the availability of other time-slots. Information flow is not continuous for any user, but rather is sent and received in bursts. The bursts are re-assembled at the receiving end and appear to provide continuous sound because the process is very fast.

Figure 16.14 shows the principle of operation. Each row in the Figure 16.14 represents a single channel and divided into three time-slots. Calls in a TDMA system start in analogue format and are sampled, transforming the call into a digital format. After the call is converted into digital format, the TDMA system places the call into an assigned time slot.

TDD: In cellular communications, when a user moves from one cell to another there is a chance that user could experience a call loss if there are no free time slots available. TDMA uses different time slots for transmission and reception. This type of duplexing is referred to as time division duplexing (TDD). TDD does not require duplexers. From Figure 16.15 TDD provides two simplex time slots on the same frequency.

Number of users supported by TDMA illustrates the improvement in the efficiency of TDMA system; let us compare Figures 16.8 and 16.13. It is obvious that the FDMA system (Figure 16.8)

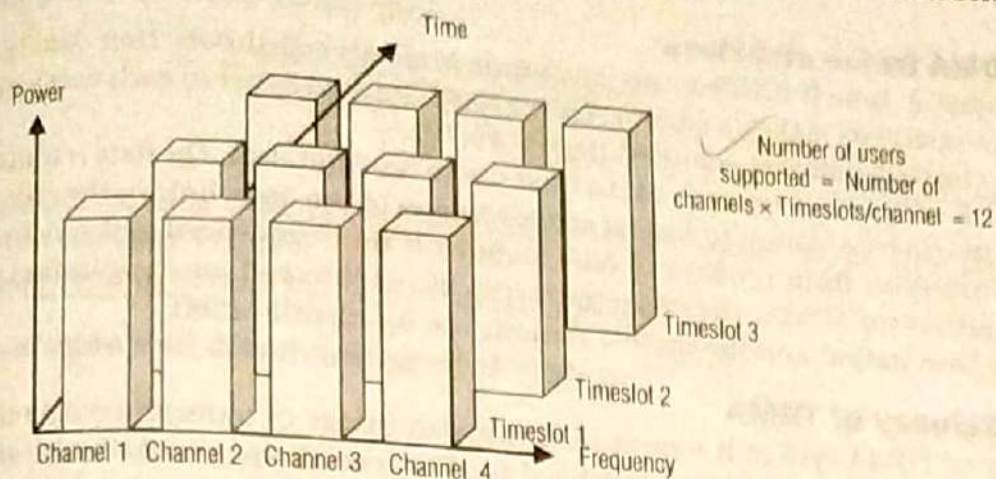


Figure 16.14 TDMA principle of operation

supports 4 users while the TDMA system (Figure 16.14) supports 12 users within the same bandwidth as the FDMA system (Equation (16.3)).

$$\begin{aligned} \text{Number of users supported by the TDMA system} \\ &= \text{Number of channels in the frequency spectrum} \times \text{Time slots/channel} \\ &= 4 \times 3 = 12 \text{ (from Figure 16.14)} \end{aligned} \quad (16.4)$$

Nowadays, there are systems in place that allow an increase of up to six times the capacity of FDMA alone. Because TDMA systems also split an allotted portion of the frequency spectrum into smaller slots (channels), they require the same level of frequency planning as FDMA systems. The same careful steps in frequency planning must be taken in both FDMA and TDMA systems.

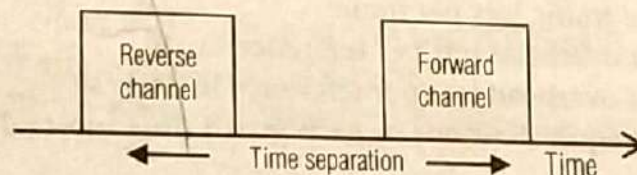


Figure 16.15 TDD two simplex time slots

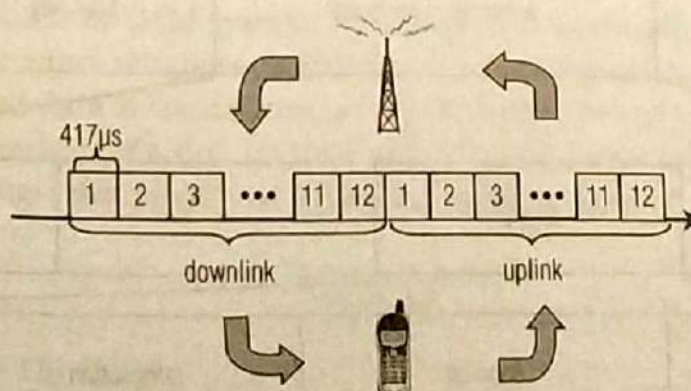


Figure 16.16 TDMA/TDD-example

16.4.2 TDMA frame structure

In a TDMA system, time is divided into equal time intervals called *slots*. User data is transmitted in the slots. Several slots make up a frame. Guard times are used between each user's transmissions to minimize crosstalk between channels (Figure 16.17).

Each user is assigned a frequency and a time slot to transmit data. The data is transmitted via a radio-carrier from a base station to several active mobiles in the downlink. In the reverse direction (uplink), transmission from mobiles to base stations is time-sequenced and synchronized on a common frequency for TDMA. The preamble carries the address and synchronization information that both the base station and the mobile stations use for identification.

16.4.3 Efficiency of TDMA

The efficiency of TDMA system is a measure of the percentage of transmitted data that contains information as opposed to providing overhead for the access scheme. The frame efficiency η is the percentage of bits per frame which contains transmitted data.

$$\begin{aligned} \text{Efficiency } (\eta) &= \frac{\text{Number of bits per frame containing in the transmitted data}}{\text{Total number of bits per frame}} \quad (16.5) \\ &= \frac{(b_T - b_{OH})}{b_T} \times 100 = (1 - b_{OH} / b_T) \times 100 \end{aligned}$$

where

b_T is the total number of bits per frame $= T_f \times R$ (16.6)

T_f is the frame duration

R is the channel bit rate

b_{OH} is the number of overhead bits per frame $= N_r \times b_r + N_t \times b_p + N_t \times b_s + N_r \times b_g$ (16.7)

where

N_r is the Number of reference bits per frame

N_t is the number of traffic bits per frame

b_r is the number of overhead bits per reference burst

b_p is the number of overhead bits per reference in each slot

b_g is the number of equivalent bits in each guard time interval

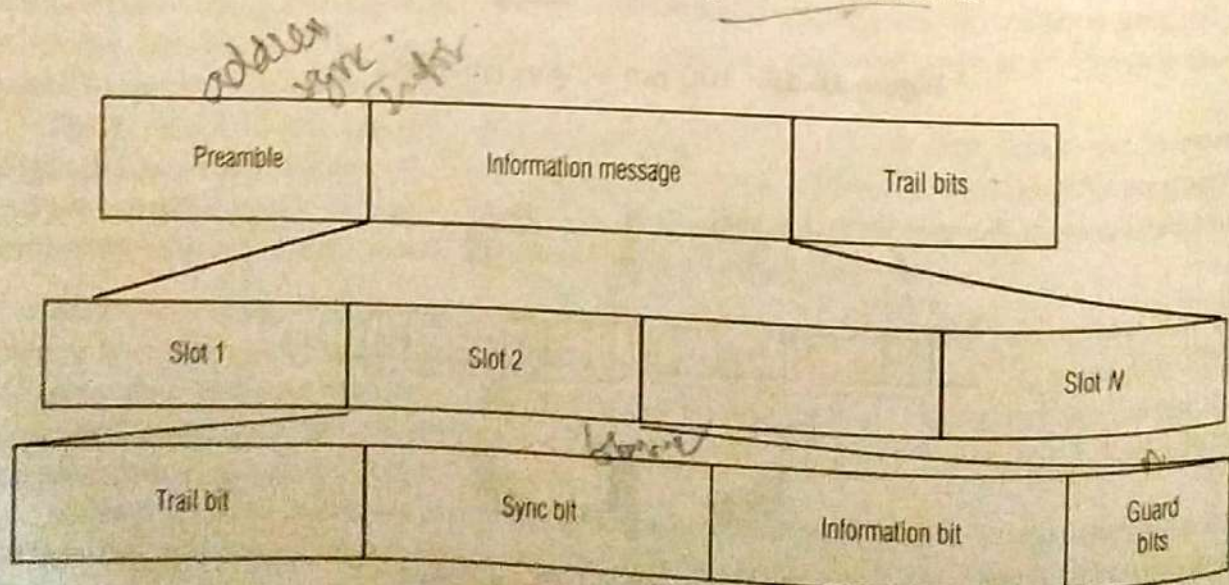


Figure 16.17 TDMA frame structure

Number of channels/time slots in TDMA system

$$N = \frac{M(B_{\text{total}} - 2B_{\text{guard}})}{B_c} \quad (16.8)$$

where

M is the number of time slots per carrier channel or maximum number of TDMA users supported on each radio channel

B_g is the guard band to prevent user at the edge of the band

16.4.4 Advantages and disadvantages of TDMA

Advantages

- Data Transmission is in discrete bursts
 - Extended battery life over FDMA and talktime
 - Handoff process is simpler, since it is able to listen for other base stations during idle time slots
- More efficient use of spectrum, compared to FDMA
 - Will accommodate more users in the same spectrum space than an FDMA system which improves capacity in high-traffic areas, such as large metropolitan areas
 - Efficient utilization of hierarchical cell structures – pico, micro, and macro cells
- Since different slots are used for transmission and reception, duplexers are not required

Disadvantages

- TDMA requires synchronization. If the time slot synchronization is lost, the channels may collide with each other.
- For mobiles and, particularly for handsets, TDMA on the uplink demands high-peak power in transmit mode that shortens battery life. If a TDMA interface consists of n channels, then the transmitted powers are $10 \log n$ times higher than in an FDMA system.
- Network and spectrum planning are intensive.
- Dropped calls are possible when users switch in and out of different cells.
- Higher costs due to greater equipment sophistication.
- Equalization is required, since transmission rates are generally very high as compared to FDMA channels.

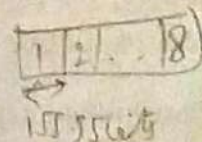
Example problem 16.3

In a digital cellular system such as GSM system, TDMA is used as multiple access system. The GSM system uses a frame structure where each frame consists of eight time slots, and each time slot contains 155.55 bits, and data is transmitted at 270.833 Kbps in the channel. Find (a) time duration of a bit, (b) time duration of a slot, (c) time duration of a frame, and (d) the duration for which a user occupying a single slot must wait between two simultaneous transmissions.

Solution

$$\begin{aligned} \text{Time duration of a bit} &= T_b = 1/\text{bit rate} \\ &= 1/270.833 \times 10^3 = 3.692 \mu\text{s} \end{aligned}$$

$$\text{Time duration of a slot} = T_{\text{slot}} = 155.55 \times T_b = 0.574 \text{ ms}$$



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Time duration of a frame $8 \times T_{\text{slot}} = 4.592 \text{ ms}$
User has to wait 4.592 ms before next transmission

Example problem 16.4

If a normal GSM timeslot consists of 6 trailing bits, 8.25 guard bits, 26 training bits, and 2 traffic bursts of 61 bits of data, find the frame efficiency

Solution

Number of bits in a time slots = $6 + 8.25 + 26 + 2(61) = 162.25 \text{ bits}$.

Number of bits in Frame = $8 \times 162.25 = 1,298 \text{ bits/frame}$.

The number of overhead bits per frame is given by

$$b_{\text{OH}} = 8(6) + 8(8.25) + 8(26) = 322 \text{ bits}$$

Frame efficiency, $\eta = (1,298 - 322)/1,298 \times 100 = 75.192 \text{ per cent}$

Example problem 16.5

Consider a GSM system, which is in TDMA/FDD system that uses the 30 MHz forward link, which is broken into number of channels of 240 kHz. If eight speech channels are supported on a single radio channel and if no guard band is assumed, find the number of simultaneous users that can be accommodated in GSM.

Solution

The number of simultaneous users that can be accommodated in GSM is

$$N = \frac{30 \times 10^6}{(240 \times 10^3 / 8)} = 1,000$$

Thus, GSM can accommodate 1,000 simultaneous users.

Example problem 16.6

If normal GSM time slot consists of 6 trailing bits, 8.50 guard bits, 28 training bits, and 2 traffic burst of 58 bits of data, find the frame efficiency.

Solution

A time slot has $6 + 8.50 + 28 + 2(58) = 158.5 \text{ bits}$

A frame has $8 \times 158.50 = 1268 \text{ bits/frame}$

The number of overhead bits per frame is given by $b_{\text{OH}} = 8(6) + 8(8.5) + 8(28) = 340 \text{ bits}$

The frame efficiency $\eta = \left(1 - \frac{340}{1268}\right) \times 100 = 73.19\%$

16.5 FDMA versus TDMA

- If an FDMA channel is not in use, it is idle and cannot be used by other users. After the assignment, the reverse and forward channels may transmit simultaneously and continuously.
- FDMA is implemented as single channel per carrier (SCPC), and is narrow band (NB) 30 kHz.
- The symbol time is large as compared to average delay spread. Inter-symbol interference (ISI) is low, and no equalization is required for NB-FDMA.
- FDMA mobile system is less complex as compared to TDMA. Due to advancement in DSP this is changing.
- FDMA is continuous transmission scheme, so fewer overhead bits are required as compared to TDMA.
- FDMA systems have higher cell site costs as compared to TDMA, because of SCPC in FDMA, and use of expensive pass band filters to eliminate the spurious radiations at base stations.
- The FDMA mobile unit uses duplexers, since both transmitter and receiver operate at the same time, resulting in an increase in the cost of mobile units and base stations.
- FDMA requires tight RF filtering to minimize adjacent channel interference.

16.6 Code division multiple access

CDMA is the third multiple access technique used in cellular systems. CDMA allows transmissions to occupy the entire bandwidth at the same time without interference.

CDMA uses spread-spectrum technique to increase spectrum efficiency over current FDMA and TDMA systems.

A spread-spectrum signal is a signal that has an extra modulation that expands the signal bandwidth beyond what is required by the underlying data modulation. Spread-spectrum communication systems are useful for the following:

1. Suppressing interference
2. Making interception difficult
3. Accommodating fading and multipath channels
4. Providing a multiple-access capability

The most practical and dominant methods of spread-spectrum communications are direct-sequence modulation and frequency hopping of digital communications. These aspects, which are basis for the CDMA, are discussed in detail in under spread-spectrum techniques in Chapter 15.

CDMA cellular technology is originally known as IS-95, which competes with GSM technology for dominance in the cellular world. CDMA cellular systems operate in the 800 MHz and 1.9 GHz PCS bands. QUALCOMM is the developer of the CDMA air interface used in cellular systems. Compared to GSM cellular systems, CDMA requires fewer cell towers and provides up to five times the calling capacity. CDMA also provides more than 10 times the voice traffic of earlier analogue system (AMPS) and is the basis for 3G data transmission for GSM carriers.

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16.6.1 CDMA principle of operation

CDMA uses unique spreading codes to spread the base band data before transmission.

CDMA assigns to each user a unique code sequence that is used to code data before transmission. If a receiver knows the code sequence related to a user, it is able to decode the received data.

The codes are shared by the mobile phone and the base station. The codes are called *pseudo-random code sequences*. All the users can access the entire spectrum allocation all of the time (Figure 16.18). A user's unique code separates the call from all other calls. The signal is transmitted in a channel, which is below noise level. The receiver then uses a correlator to despread the wanted signal, which is passed through a *narrowband pass filter*. Unwanted signals will not be despread and will not pass through the filter. Codes take the form of a carefully designed *one/zeros* sequence produced at a much higher rate than that of the base band data. The rate of a spreading code is referred to as chip rate rather than bit rate.

CDMA channels can handle an unspecified number of users. The capacity of the system depends on the quality of current calls. As more users are added, noise is added to the wideband frequency, therefore decreasing the quality of current calls. Each user's transmission power increases the level of the frequency spectrum's "noise floor," and therefore decreases the overall call quality for all users. To eliminate the "noise floor," CDMA mobile phones and base stations use the minimum amount of power required to communicate with each other. They use precise power control to decrease users' transmission power. By decreasing a user's transmission power, the mobile phone has added battery life, increased talktime, and smaller batteries.

Types of codes used in CDMA: There are three types of codes generally used. They are Walsh code, Short PRN code, and Long PRN codes.

Walsh codes: These are orthogonal codes. The spreading on forward link is 1.2288 Mbps and on reverse link is 307.2 Kbps. 64-bit Walsh codes are used in IS 95A and IS 95B. 128-bit Walsh codes are used in CDMA2000.

Short PRN code: (16 bit) are used to identify the base station and the cell.

Long PRN code : (42-bit code) are used to identify mobile station on reverse link.

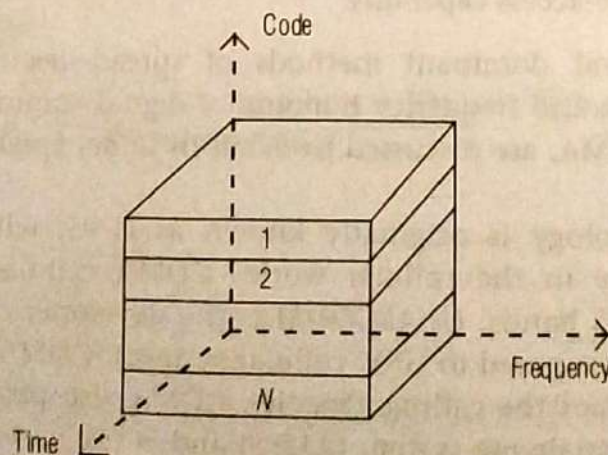


Figure 16.18 CDMA users access entire spectrum

16.6.2 Advantages and disadvantages of CDMA**Advantages**

- Greatest spectrum efficiency: capacity increases about 8 to 10 times that of an analogue system and 4 to 5 times that of other digital systems, which makes it most useful in high traffic areas with a large number of users and limited spectrum.
- CDMA improves call quality by filtering out background noise, crosstalk, and interference.
- "Soft handoffs": because of the multiple diversities in use, handoffs between cells are undetected by the user.
- Simplified frequency planning: all users on a CDMA system use the same radio frequency spectrum.
 - Engineering detailed frequency plans are not necessary.
 - Frequency re-tunes for expansion are eliminated.
 - Fewer cells are required for quality coverage.
- Random Walsh codes enhance user privacy; a spread-spectrum advantage.
- Precise power control increases talktime and battery life for mobile phones.

Disadvantages

- Backwards compatibility techniques are costly.
- Currently, equipment is expensive.
- Difficult to optimize to maximize performance.
- Low traffic areas lead to inefficient use of spectrum and equipment resources.

16.6.3 Handoffs in CDMA mobile systems

The act of transferring a call of from one base station to another is termed as handoff. Handoff occurs when a call has to be handed off from one cell to another as the user moves between cells.

Hard handoff and soft handoff

In a traditional "hard" handoff, the connection to the current cell is broken and then the connection to the new cell is made. This is known as a "break-before-make" or hard handoff. Since all cells in CDMA use the same frequency, it is possible to make the connection to the new cell before leaving the current cell. This is known as a "make-before-break" or "soft handoff". Soft handoff requires less power, which reduces interference and increases capacity. The implementation of handoff is different in GSM and CDMA standards.

Differences between soft handoff and hard handoff

- Soft handoff is diverse from the traditional hard handoff process.
- Handoff is introduced and performed without the user attempting to have synchronized the traffic channel communications with the two base stations.
- With the soft handoff, a conditional decision is build whether to handoff or not. Depending on the changes in pilot signal strength from the two or more base stations included, a hard decision will ultimately be made to communicate with only one base station. This is normally happens after it is apparent that the signal from one base station is significantly stronger than those from the others. Temporarily, the user has simultaneous traffic channel communication with all candidate base stations.
- With hard handoff, a definite decision has to be made on whether to handoff or not.

Soft handoff example

A simple case of soft handoff is shown in Figure 16.19. There are only two base stations involved in the example. The same basic setup has two outcomes, one of which is shown in the left column as "scenario one," and the other in the right column as "scenario two". Scenario one shows a soft handoff from BS1 to BS2. Scenario two shows a case in which the user goes back to BS1 after a period of time in soft handoff. The equivalent scenario with hard handoff would be a hard handoff to BS2 and then another hard handoff back to BS1, wasting valuable network resources in the process of carrying out these unnecessary handoffs.

From Figure 16.19: (a) The left column shows a user at first having only BS1 in its active set; (b) Followed by a period time in which the pilot signal strengths of both BS1 and BS2 are strong, so the user has both BS1 and BS2 in its active set; (c) Eventually, the signal strength of BS1 declines and the signal strength of BS2 increase to the point where BS1 is removed from the active set. The column on the right shows the same situation, except that at the end BS2 is removed from the active set.

More detailed description on soft handoffs in CDMA mobile systems is given under CDMA mobile systems in Chapter 18.

16.6.4 Multipath error elimination in CDMA cellular system

One of the main advantages of CDMA systems is the capability of using signals that arrive in the receivers with different time delays. This phenomenon is called multipath.

FDMA and TDMA, which are narrowband systems, cannot discriminate between the multipath arrivals, and resort to equalization to mitigate the negative effects of multipath. Due to its wide bandwidth and rake receivers, CDMA uses the multipath signals and combines them to make a stronger signal at the receivers. One of the receivers (fingers) constantly search for different multipath and feeds the information to the other three receivers. Each receiver then

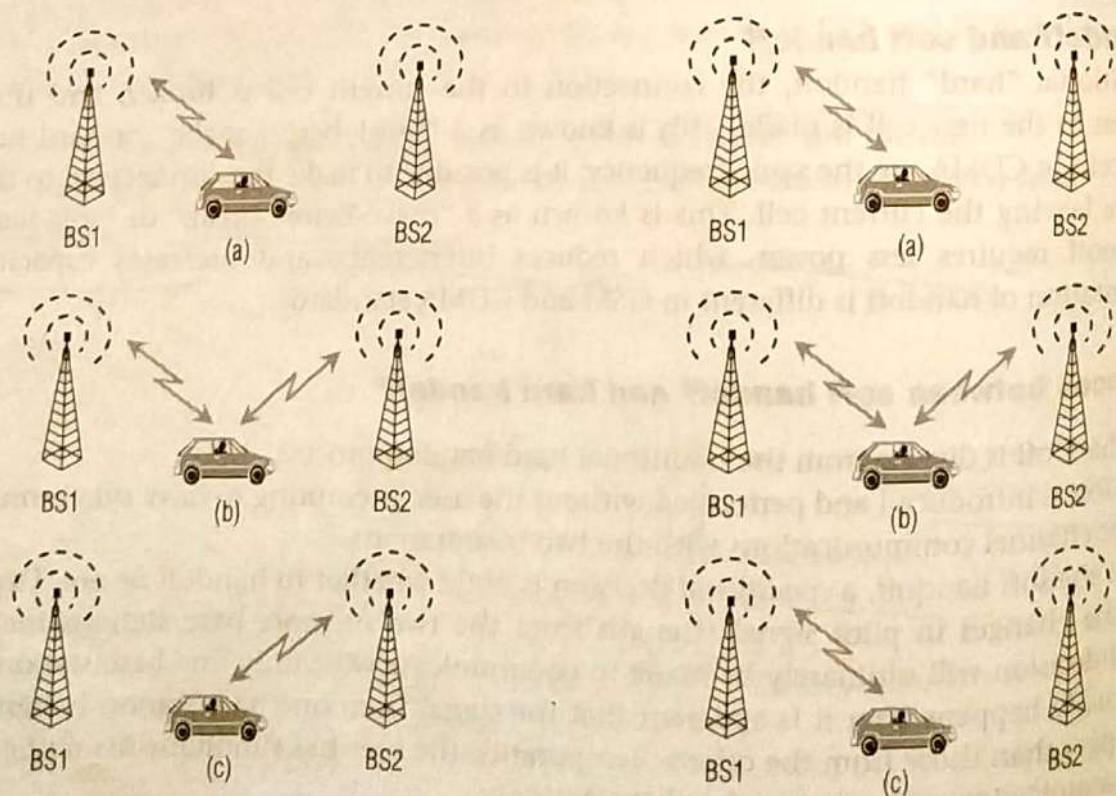


Figure 16.19 A simple soft handoff situation with two different outcomes

demodulates the signal corresponding to a strong multipath. The results are then combined together to make the signal stronger.

16.6.5 Comparison of CDMA- and TDMA/FDMA-based cellular systems

- In FDMA, all the users occupy disjoint frequency bands but are transmitted simultaneously in time.
- In TDMA, whereby all users occupy the same bandwidth but transmit in disjoint intervals of time.
- In CDMA, all the signals occupy the same bandwidth and are transmitted simultaneously in time. The different waveforms in CDMA are distinguished from one another at the receiver by the specific spreading codes they employ. It utilizes the most important use of spread-spectrum communications as a multiple accessing technique.
- In TDMA/FDMA, cell design requires more frequency planning which is tough job. Whereas in CDMA frequency planning is minimal.
- In TDMA, bandwidth available for transmission is small which leads to compromise in quality of transmission. Whereas in CDMA systems entire spectrum is used, this enhances voice quality.
- TDMA is band-limited system. CDMA is power-limited system.
- CDMA popularity is primarily due to the performance of the spread-spectrum waveforms display when transmitted over a multipath fading channel.

16.7 Near-far problem

A common difficulty encountered in CDMA wireless communication systems implemented using direct sequence spread spectrum (DSSS) technology is referred to as the near-far problem. The near-far problem occurs when two or more DSSS transmitters transmit the signals towards the same DSSS receiver as shown in Figure 16.20. In this figure, two DSSS transmitters transmit the signals towards a DSSS receiver, one transmitter being closer to the receiver than the other transmitter. The power of the two spread-spectrum signals transmitter is the same at the antenna of each DSSS transmitter.

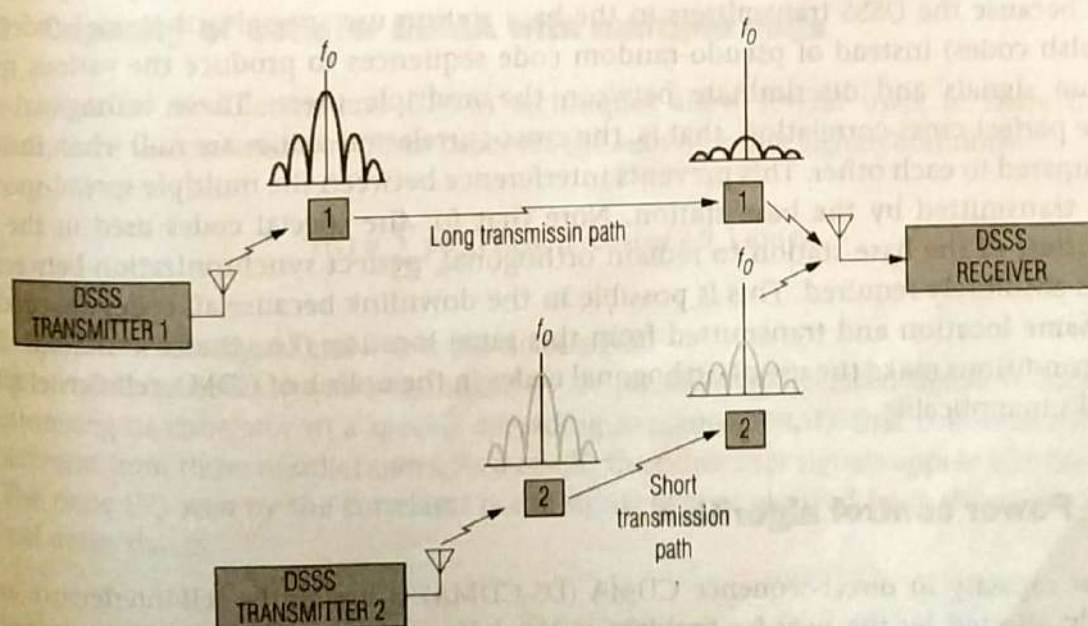


Figure 16.20 The near-far problem

However, the two spread-spectrum signals at the DSSS receiver antenna have different power levels because the paths between the two transmitters and the receiver are of different lengths. In Figure 16.20, the power of the spread-spectrum signal coming from DSSS transmitter 1 is lower than that coming from DSSS transmitter 2 because transmitter 1 is farther away from the receiver than transmitter 2. This can be a serious problem when the DSSS receiver is set to demodulate the spread-spectrum signal coming from DSSS transmitter 1, because the power level of this signal is lower than that of the spread-spectrum signal coming from DSSS transmitter 2. Since any spread-spectrum signal other than the desired one produces interference similar to that caused by noise, this results in a rather poor S/N ratio at the DSSS receiver input. Consequently, errors are likely to appear in the recovered data when the process gain of the system is not sufficient to overcome the S/N ratio deficit observed at the DSSS receiver input.

16.7.1 Near-far problem in the uplink of CDMA

The near-far problem described above is encountered in the CDMA cellular-telephone uplink networks. In such networks, each cellular telephone set is, in fact, a DSSS transmitter that transmits a spread-spectrum signal towards the base station (DSSS receiver). The power level of each spread-spectrum signal received depends on the distance that separates the corresponding cellular telephone set from the base station. The near-far problem can be greatly mitigated by having each cellular telephone set (DSSS transmitter) make its output signal power proportional to the distance that separates it from the base station. This causes the cellular telephone sets located close to the base station to reduce their output signal power. Conversely, the cellular telephone sets that are far from the base station boost their output signal power. The net effect is that the power levels of all spread-spectrum signals received at the base station are approximately the same. Power control in cellular telephone sets also offers another great benefit: it reduces the average energy consumption, thereby increasing the time during which the unit can be used before the battery needs to be recharged.

16.7.2 Near-far problem in the downlink of CDMA

The near-far problem does not happen in the downlink of CDMA cellular telephony networks. This is because the DSSS transmitters in the base station use special orthogonal codes (e.g., the Walsh codes) instead of pseudo-random code sequences to produce the various spread-spectrum signals and discriminate between the multiple users. These orthogonal codes provide perfect cross-correlation, that is, the cross-correlation values are null when the codes are compared to each other. This prevents interference between the multiple spread-spectrum signals transmitted by the base station. Note that for the special codes used in the DSSS transmitters of the base station to remain orthogonal, perfect synchronization between the codes is absolutely required. This is possible in the downlink because all codes are produced at the same location and transmitted from this same location (i.e., the base station). These severe conditions make the use of orthogonal codes in the uplink of CDMA cellular-telephone networks inapplicable.

16.8 Power control algorithm

The user capacity in direct-sequence CDMA (DS-CDMA) is limited by self-interference and is adversely affected by the near-far problem at the base station receiver. Thus, accurate power control of all the mobile radio transmitters in the system is essential and an added challenge

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for the transceiver design. The receiver includes an automatic gain control (AGC) loop to track the received power level, which varies because of large-scale path loss and small-scale fading. To compensate for those effects, CDMA (or IS95) employs two power control methods:

- Open-loop power control
- Closed-loop power control

Open-loop power control: The open loop method uses the power level at the mobile radio receiver (P_{Rx}) to estimate the forward-link path loss. It then specifies the transmit power (P_{Tx}) of the mobile radio as

$$P_{Tx} = -73 \text{ dBm} - P_{Rx} \quad (16.9)$$

Example problem 16.7

If the received power level is -85 dBm , find the transmitted power level.

Solution

$$\begin{aligned} \text{Received power } P_{Rx} &= -85 \text{ dBm} \\ \text{Transmit power level } P_{Tx} &= (-73 \text{ dB}) - (-85 \text{ dB}) = +12 \text{ dBm} \end{aligned}$$

Closed-loop power control: Adding a feedback signal completes the AGC loop and improves the accuracy of the open-loop method. The feedback signal is an error signal sent from the base station to the mobile radio that instructs the mobile radio to increase or decrease power by a set amount, generally 1 dB . It is sent once per power control group and is therefore updated at a rate of **800 Hz**. As such, it is sufficient to support vehicle speeds up to 100 km/h . This second power control method is referred to as closed-loop power control.

16.11 Space division multiple access

In addition to frequency, time, and code domains, the spatial dimension can also be used for multiplexing of different data streams by transmitting the data streams over different,

non-overlapping transmission channels. The use of space division multiplexing for multiple access is termed SDMA.

SDMA enables users to share simultaneously the same bandwidth in different geographical locations. SDMA solves capacity problem of wireless communication systems by exploitation of the spatial dimension which makes it possible to identify the individual users, even when they are in the same time/frequency/code domains. SDMA can be achieved using beam forming or sectorization.

① Omnidirectional antenna-based traditional cellular networks

In traditional mobile cellular network systems, the base stations have no information on the position of the mobile units within the cell, and base stations radiate the signal in all directions within the cell in order to provide radio coverage. These results in wasting power on retransmissions when there are no mobile units to reach, in addition to causing interference for adjacent cells using the same frequency, so called co-channel cells. In the same way, in reception, the antenna receives signals coming from all directions including noise and interference. These considerations have led to the development of the SDMA technique, which is based on deriving and exploiting information on the spatial position of mobile terminals.

Adaptive antenna or SDMA-based cellular network

By using adaptive antennas arrays, sometimes called smart antennas in mobile radio systems, signals can be received and sent only from and into a limited angular range, following the directional nature of multipath. This improves coverage or link quality in noise-limited situations and enhances capacity in interference-limited situations. The concept of SDMA is shown in Figure 16.22. Each user exploiting a single-transmitter-antenna-aided mobile station simultaneously communicates with the base station equipped with an array of receiver antennas.

16.11.1 Antenna arrays

By combining the outputs of the individual antennas in an array, a single effective antenna can be created with gain and directional characteristics that are very different from those of the individual elements comprising the array. For example, consider a row of, m , simple and

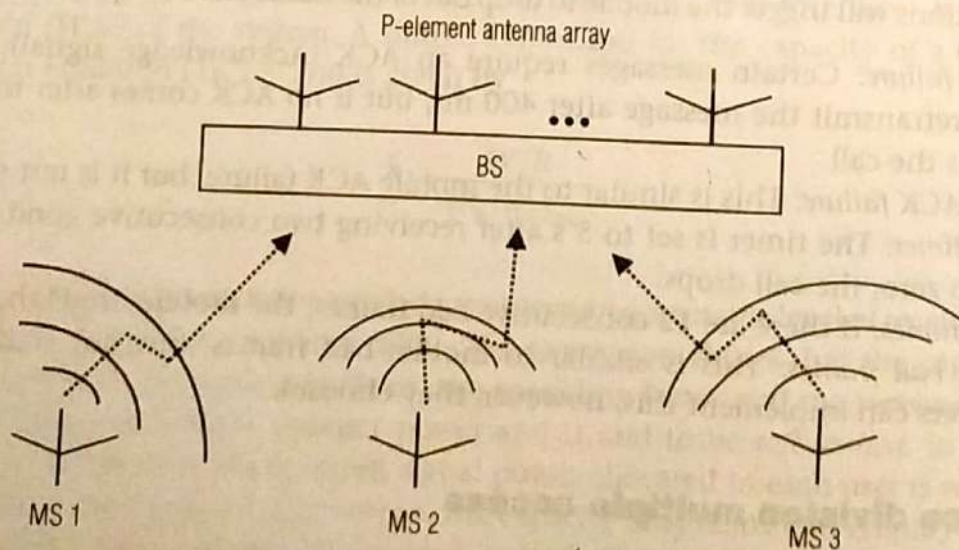


Figure 16.22 SDMA concept, employing a P-element receiver antenna array for supporting three mobile users

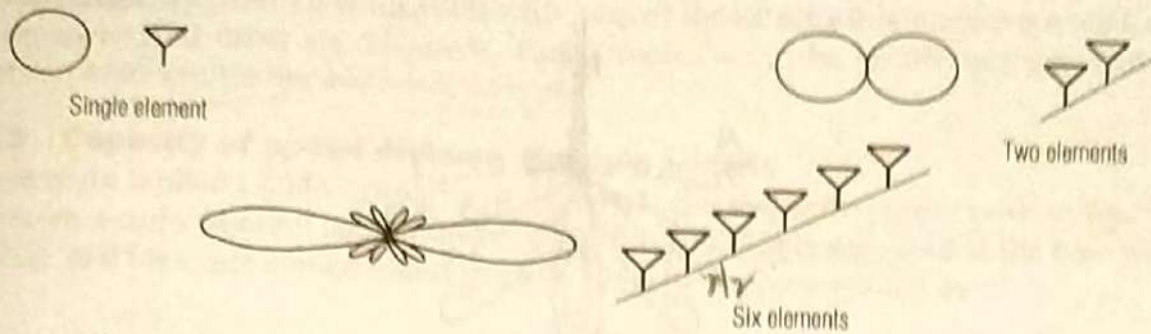


Figure 16.23 Antenna patterns for arrays of different number of elements

identical, antenna elements (called a linear array) with elements spaced one-half wavelength apart. Suppose that one simply adds the outputs of all the elements. Signals arriving at the array from broadside, or perpendicular to the axis of the array, arrive simultaneously at each element of the array, and their sum will therefore be m times as large (m^2 times as powerful) as the signal received by a single antenna. Simply adding the outputs of the m element array therefore results in an amplitude gain of m for signals arriving from broadside.

Figure 16.23 depicts the effective antenna pattern resulting from the above strategy for linear arrays consisting of a single antenna, two antennas, and six antennas (omnidirectional element patterns and half-wavelength spacing in all cases). The radius of the pattern in Figure 16.23 is proportional to the gain or strength of the signal at the output of the array. The increased gain in the broadside direction and the increases in gain with the number of array elements are evident in the Figure 16.23. It is also evident that there are certain directions in which the effective antenna has reduced sensitivity, or nulls.

Classical phased-array antenna systems have gain patterns or beam patterns that are shown in the figure, except that the direction of maximum gain might not be the broadside direction. More advanced systems have patterns that are optimized to enhance a particular user's signal while simultaneously rejecting interferers. Another differentiator of antenna array systems is whether or not the beam patterns can change with time. Antenna arrays can be built with combining strategies that are fixed. From an operational standpoint, this sort of array is no different from a conventional antenna with the same directional sensitivity. Alternatively, arrays can be equipped with combining hardware and software that make it possible for the pattern to be changed over time and adapted to the current operational scenario. Because radio reception and transmission are reciprocal, directive transmission with gain is also possible from an array. Any directivity pattern achievable for reception is also achievable for transmission.

SDMA controls the radiated energy for each user in space (Figure 16.24) by using adaptive antenna array patterns. These different areas covered by the antenna beam may be served by the same frequency (in a TDMA or CDMA system) or different frequencies (in an FDMA system). Sectorized antennas may be thought of as a primitive application of SDMA. Figure 16.25 shows spatially filtered base station antenna serving different users by using spot beams.

16.11.2 Advantages of SDMA technique

The advantages of SDMA system include the following:

1. **Range extension:** The coverage area of the antenna array is greater than that of any single element as a result of the gain provided by the array. When a system is constructed using SDMA, the number of cells required to cover a given area can be substantially reduced.

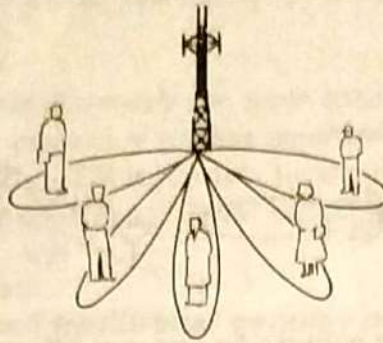


Figure 16.24 Multiple users belonging to the same cell use the same channel

A 10-element array offers a gain of 10, which typically doubles the range of the cell and thereby quadruples the coverage area.

2. **Interference suppression:** Interference from other systems and from users in other cells is significantly reduced by exploiting the desired user's unique channel impulse responses (CIRs). In "noisy" areas where range is limited by interference, spatially selective transmission and reception result in range extension.
3. **Multipath effect elimination:** The copies of the desired signal that have arrived at the antenna after bouncing from objects between the signal source and the antenna can often be mitigated. In certain cases, the multipath can actually be used to reinforce the desired signal.
4. **Capacity increase:** Capacity increase can be done in two ways:
 - Channel reuse patterns in cellular systems can be significantly tighter because the average interference resulting from co-channel signals in other cells is markedly reduced (e.g., moving from a 7-cell to a 4-cell reuse pattern nearly doubles capacity).
 - Separate spatial channels can be created in each cell on the same conventional channel. In other words, intracellular reuse of conventional channels is possible (Figure 16.25).

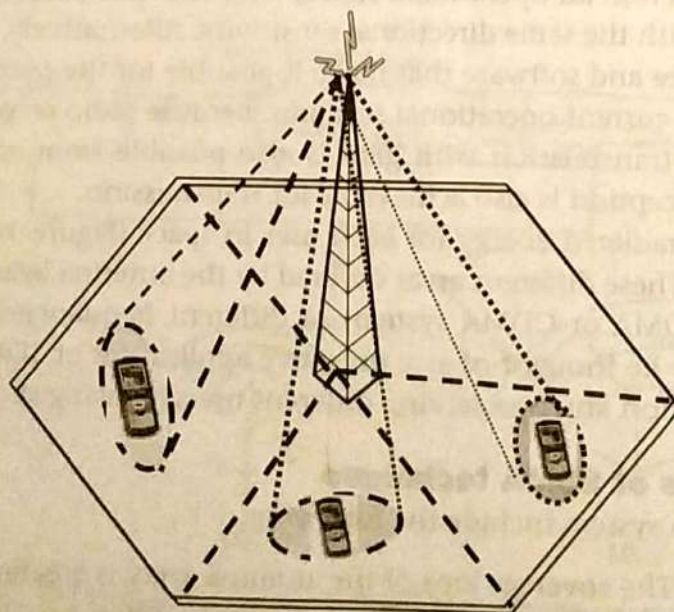


Figure 16.25 Intra-cell SDMA

5. **Compatibility:** SDMA is compatible with most of the existing modulation schemes, carrier frequencies and other specifications. Furthermore, it can be readily implemented using various array geometries and antenna types.

16.11.3 Capacity of space division multiple access

For interference limited CDMA operating in an AWGN channel, with perfect power control with no interference from adjacent cells and with omnidirectional antennas used at the base stations, the average BER for a user can be found from the Gaussian approximation as

$$P_b = Q\left(\sqrt{\frac{3N}{K-1}}\right) \quad (16.16)$$

where K is the number of users in a cell and N is the spreading factor. $Q(x)$ is the standard Q-function.

The directive antennas can improve the reverse link in a single-cell CDMA system. The omnidirectional receiver antenna will detect signals from all users in the system, and thus will receive the greatest amount of noise. The sectorized antenna will divide the received noise into a smaller value and will increase the number of users in the CDMA system. A spot beam for each user, and it is this implementation which is the most powerful form of SDMA. An ideal adaptive antenna is able to form a beam for each user in the cell of interest, and the base station tracks each user in the cell as it moves. Assume that a beam pattern, is formed such that the pattern has maximum gain in the direction of the desired user. Such a directive pattern can be formed at the base station using an N-element adaptive array antenna.

16.11.4 Comparison of SDMA, TDMA, FDMA, and CDMA

The performances of the various multiple access techniques are compared and shown in Table 16.2.

16.12 Orthogonal frequency division multiplexing

OFDM is an important technology for all next generation wireless systems (e.g., 4G system, HiperLAN) as it offers numerous advantages over other existing technologies, such as robust performance over multipath fading channels and the ability to achieve high spectral efficiency.

OFDM is a form of multi-carrier modulation where high-rate data stream is split into a number of lower rate streams that are transmitted simultaneously over a number of subcarriers (Figure 16.26).

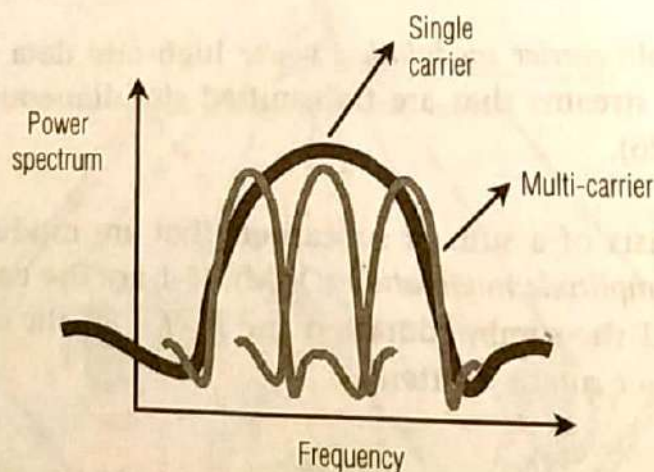
An OFDM signal consists of a sum of subcarriers that are modulated by using phase shift keying (PSK) or quadrature amplitude modulation (QAM). If d_i are the complex QAM symbol, N_s is the number of subcarriers, T the symbol duration and $f_i = f_0 + i/T$ the carrier frequency, then one OFDM symbol starting at $t = t_s$ can be written as:

$$s(t) = \text{Re} \left\{ \sum_{i=0}^{N_s-1} d_i \exp(j2\pi f_i(t-t_s)) \right\} \quad t_s \leq t \leq t_s + T$$

$$s(t) = 0, \quad t < t_s \wedge t > t_s + T \quad (16.17)$$

Table 16.2 Comparison of SDMA, TDMA, FDMA, and CDMA

Approach	SDMA	TDMA	FDMA	CDMA
Idea	Segment space into cells/sectors	Segment sending time into disjoint time slots, demand driven or fixed patterns	Segment the frequency band into disjoint sub-bands	Spread the spectrum using orthogonal codes
Terminals	Only one terminal can be active in one cell/one sector	All the terminals are active for short periods of time on the same frequency	Every terminal has its own frequency, uninterrupted	All terminals can be active at the same place at the same moment, uninterrupted
Signal separation	Cell structure directed antennas	Synchronization in the time domain	Filtering in the frequency domain	Code plus special receivers
Advantages	Very simple, increases capacity per km ²	Established fully digital, very flexible	Simple established, robust	Flexible, less planning needed, soft handover
Disadvantages	Inflexible antennas typically fixed	Guard space needed (multi-path propagation), synchronization difficult	Inflexible frequencies are as scarce resource	Complex receivers, needs more complicated power control for senders
Comment	Only combination with TDMA, FDMA, or CDMA useful	Standard in fixed networks, together with FDMA/SDMA used in many mobile networks	Typically combined with TDMA (frequency hopping patterns) and SDMA (frequency reuse)	Used in many 3G systems higher complexity, lowered expectations; integrated with TDMA/FDMA


Figure 16.26 Power spectra of single- and multi-carrier systems

By using the equivalent complex notation, Equation (16.17) can be written as

$$s(t) = \sum_{i=0}^{N_s-1} d_i \exp(j2\pi f_i(t-t_i)) \quad t_s \leq t \leq t_s + T$$

$$s(t) = 0, \quad t < t_s \wedge t > t_s + T$$

In this representation, the real and imaginary parts correspond to the in-phase and quadrature parts of the OFDM signal, which have to be multiplied by a cosine and sine of the desired carrier frequency to produce the final OFDM signal. OFDM modulator (Figure 16.27) and its equivalent circuit is shown for transmission of QAM symbols on parallel subcarriers overlapping.

The biggest advantage of the OFDM technique is the mutual orthogonality of its carriers; this provides a high spectral efficiency. This is possible because there is no guard band, and carriers can be packed very close together.

We will discuss key features of OFDM technology in this section.

16.12.1 Basic principle of operation

The concept of parallel transmission of data is the basis for OFDM technique. The basic idea of OFDM is to transmit N symbols in parallel over N different subcarriers while enlarging the symbol duration N times.

We can describe OFDM principle through a rotation of the time frequency plane as shown in Figure 16.27. In a single-carrier system, each symbol occupies the full bandwidth. In a multi-carrier system the symbol duration is enlarged N times and simultaneously the bandwidth consumption of each symbol is reduced by the same factor N .

Figure 16.28 illustrates the difference between a single-carrier and a multi-carrier system through a rotation of the time-frequency plane. The transmitted data symbols are denoted $d[1] \dots d[8]$. The multi-carrier system uses $N = 8$ subcarriers.

Figure 16.29 illustrates the subcarrier frequency-spectra in an OFDM system. The subcarrier has bandwidth Δf . The centre frequency of subcarrier q is denoted by f_q .

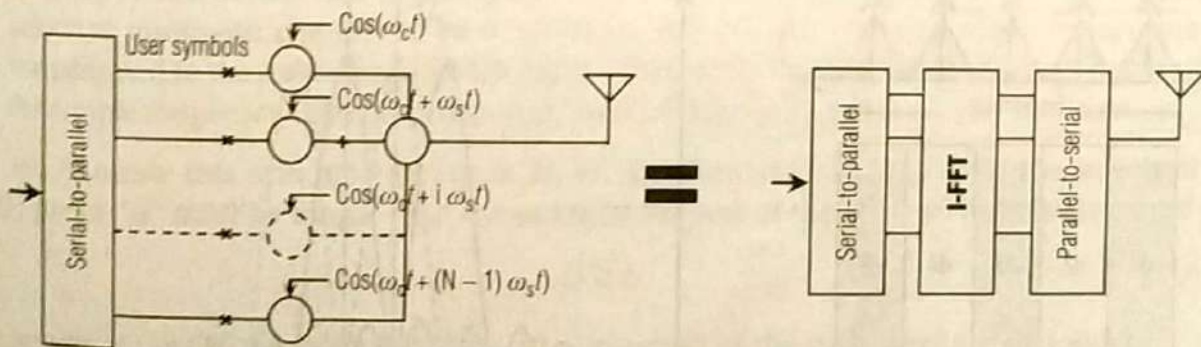


Figure 16.27 OFDM modulator

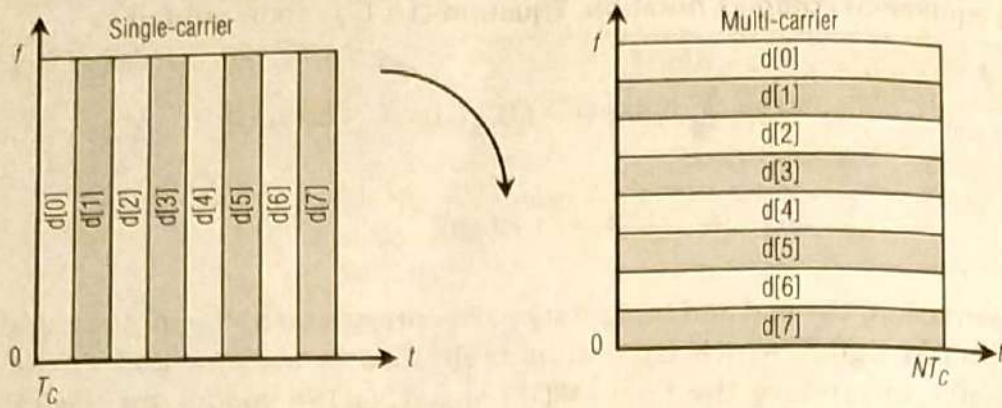


Figure 16.28 Single-carrier and multi-carrier systems

The overall data-rate and bandwidth consumption is kept constant through parallel transmission over N independent subcarriers. The subcarrier spectra overlap, as depicted in Figure 16.29. However, if the centre frequency of each subcarrier q is chosen as

$$f_q = q/(NT_C) \quad (16.18)$$

for $q \in \{0, \dots, N-1\}$ the subcarriers are orthogonal despite their overlapping spectra. OFDM is a special case of a multi-carrier scheme with overlapping but orthogonal subcarriers.

Figure 16.30 shows all operations that are necessary for OFDM. Each subcarrier is modulated by a symbol (from a binary phase shift keying (BPSK) alphabet in this example) and the resulting signals are summed up. These operations are equivalent to an inverse discrete Fourier transform (IDFT). The inverse DFT can be efficiently implemented by means of the inverse fast Fourier transform. The existence of such an efficient algorithm for the actual implementation is one major reason for the widespread application of OFDM.

OFDM needs the following processing steps (Figure 16.29):

- First, the subcarriers are multiplied by the individual data symbols, and then the resulting signals are added together.
- A copy of the signal tail is inserted at the beginning of each OFDM symbol (Figure 16.31).

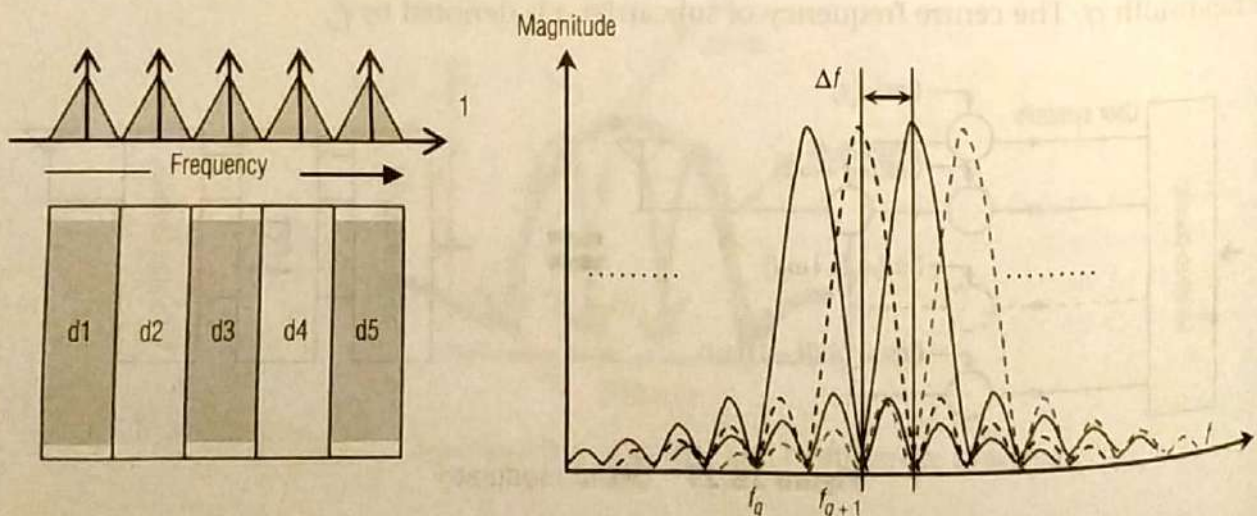


Figure 16.29 Subcarrier frequency-spectra in an OFDM system

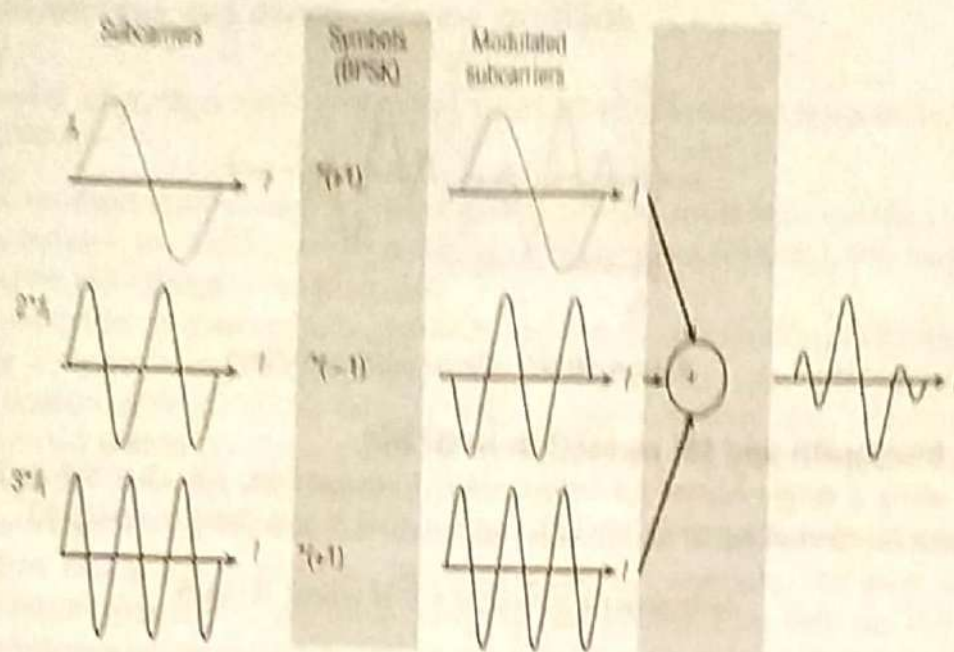


Figure 16.30 OFDM transmission

16.12.2 Cyclic prefix

In a fading multipath environment, inter-carrier interference (ICI) and ISI will be introduced. To solve this problem, every OFDM block is extended by a guard interval T_g , $T_g > T_d$ where T_d is the maximum multipath delay spread.

The most common guard interval is cyclic prefix. In order to completely remove the inter-symbol interference, a cyclic prefix is inserted in front of every OFDM symbol.

The cyclic prefix is a copy of the OFDM symbol tail.

Cyclic prefix refers to the prefixing of a symbol with a repetition of the end. Although the receiver is typically configured to discard the cyclic prefix samples, the cyclic prefix serves two purposes.

- As a guard interval, it eliminates the ISI from the previous symbol.
- As a repetition of the end of the symbol, it allows the linear convolution of a frequency-selective multipath channel to be modelled as circular convolution, which in turn may be transformed to the frequency domain using a discrete Fourier transform. This approach allows for simple frequency-domain processing, such as channel estimation and equalization.

We illustrate this operation in Figure 16.31. To eliminate ISI completely, the length of the cyclic prefix "G" must be longer than the essential support of the channel impulse response L ,

$$G \geq L \quad (16.19)$$

The length of the OFDM symbol in chips after insertion of the cyclic prefix is denoted by

$$P = N + G$$

where N is number of symbols and G is length of cyclic prefix.

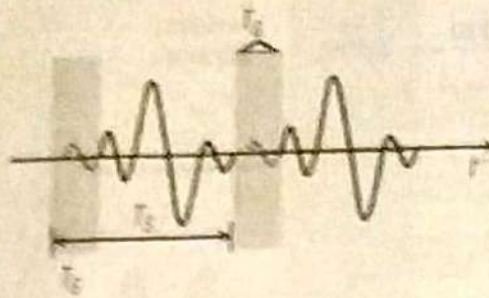


Figure 16.31 Cyclic prefix insertion

16.12.3 Multipath and ISI reduction in OFDM

Instead of using a single carrier for information transmission, the channel bandwidth " B " is divided into a number of equal bandwidth (Δf) creating N sub-channels ($N\Delta f/N$).

$$\text{Subcarrier interval } \Delta f = R/N \text{ where } B = \Delta f N$$

Consider a data stream operating at R bps and an available bandwidth of $N\Delta f$ centred at carrier frequency f_c . The entire bandwidth could be used to transmit a data stream, in which case the bit duration would be $1/R$. By splitting the data stream into N sub-streams using a serial-to-parallel converter, each sub-stream has a data rate of R/N and is transmitted on a separate subcarrier, with spacing between adjacent subcarriers of Δf (Figure 16.32). The bit duration is N/R .

The advantage of OFDM is that on a multiple channel the multipath is reduced relative to the symbol interval by a ratio of $1/N$ and thus imposes less distortion in each modulated symbol. OFDM overcomes ISI in a multipath environment. ISI has a greater impact at higher data rates because the distance between bits and symbols is smaller. With OFDM, the data rate is reduced by a factor of N , which increases the symbol duration by a factor of N . Thus, if the symbol duration is T_s for the source stream, the duration of OFDM signals is NT_s . This significantly reduces the effect of ISI.

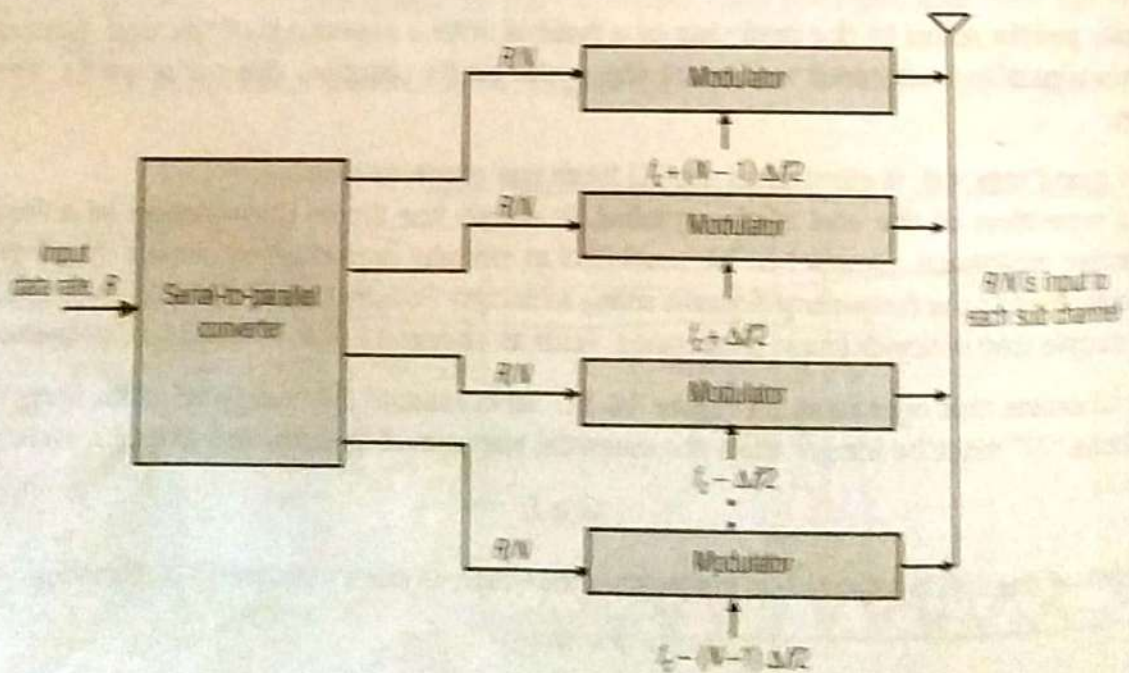


Figure 16.32 Orthogonal frequency division multiplexing (OFDM)

16.12.4 Advantages and disadvantages of OFDM

Advantages

OFDM has several advantages over other widely used wireless access techniques such as TDMA, FDMA, and CDMA.

1. In OFDM, multiple symbols can be transmitted in parallel, while maintaining a high spectral efficiency because the radio channel is divided into many narrowband, low-rate, frequency-non-selective sub-channels or subcarriers
 - i. Each subcarrier may deliver information for a different user, resulting in a simple multiple-access scheme known as orthogonal frequency-division multiple access (OFDMA). This enables different media such as video, graphics, speech, text, or other data to be transmitted within the same radio link, depending on the specific types of services and their quality-of-service (QoS) requirements.
 - ii. Furthermore, in OFDM systems different modulation schemes can be employed for different subcarriers or even for different users. For example, the users close to the base station may have a relatively good channel quality, thus they can use high-order modulation schemes to increase their data rates. By contrast, for those users who are far from the base station or are serviced in highly loaded urban areas, where the subcarriers' quality is expected to be poor, low-order modulation schemes can be invoked.

Disadvantages

1. One problem is the associated increased peak-to-average power ratio (PAPR) in comparison with single-carrier systems, requiring a large linear range for the OFDM transmitter's output amplifier.
2. In addition, OFDM is sensitive to carrier frequency offset, resulting in ICI.
3. Not ideal for mobile phones, but fine for mobile computers with bigger batteries that are not sending data continuously.
4. Transmitter and receiver must be linear to preserve shape.

16.13 Packet radio

Early radio systems transmitted analogue signals. Today most radio systems transmit digital signals composed of binary bits, where the bits are obtained directly from a data signal or by digitizing an analogue signal. A packet radio is a digital radio that can transmit a continuous bit stream or it can group the bits into packets and is characterized by bursty transmissions. The radio is idle except when it transmits a packet. Packet radio can be simply defined as technology which enables the user to transmit packet data (digital data) via radio (wireless) interface.

The first network based on packet radio, ALOHANET, was developed at the University of Hawaii in 1971. This network enabled computer sites at seven campuses spread out over four islands to communicate with a central computer on Oahu via radio transmission. The network architecture used a star topology with the central computer at its hub. Any two computers could establish a bi-directional communications link between them by going through the central hub. ALOHANET incorporated the first set of protocols for channel access and routing in packet radio systems, and many of the underlying principles in these protocols are still in use today. The U.S. military was extremely interested in the combination of packet data and broadcast radio inherent to ALOHANET.

Throughout the 1970s and early 1980s the Defense Advanced Research Projects Agency (DARPA) invested significant resources to develop networks using packet radios for tactical communications in the battlefield. The nodes in these ad hoc wireless networks had the ability to self-configure (or reconfigure) into a network without the aid of any established infrastructure. DARPA's investment in ad hoc networks peaked in the mid-1980s, but the resulting networks fell far short of expectations in terms of speed and performance. These networks continue to be developed for military use. Packet radio networks also found commercial application in supporting wide-area wireless data services.

These services, first introduced in the early 1990s, enable wireless data access (including email, file transfer, and web browsing) at fairly low speeds, on the order of 20 Kbps. A strong market for these wide-area wireless data services never really materialized, due mainly to their low-data rates, high cost, and lack of "killer applications." These services mostly disappeared in the 1990s, supplanted by the wireless data capabilities of cellular telephones and wireless local area networks (LANs).

Multiple access techniques are primarily for continuous-time applications like voice and video, where a dedicated channel facilitates good performance. However, most data users do not require continuous transmission: data is generated at random time instances, so dedicated channel assignment can be extremely inefficient. Moreover, most systems have many more total users (active plus idle users) than can be accommodated simultaneously, so at any given time channels can only be allocated to users that need them. Random access strategies are used in such systems to efficiently assign channels to the active users. All random access techniques are based on the premise of packetized data or *packet radio*. In packet radio, user data is collected into packets of N bits, and once a packet is formed it is transmitted over the channel. Assuming a fixed channel data rate of R bps, the transmission time of a packet is

$$\tau = \frac{N}{R}.$$

The transmission rate R is assumed to require the entire signal bandwidth, and all users transmit their packets over this bandwidth, with no additional coding that would allow separation of simultaneously transmitted packets. Thus, if packets from different users overlap in time a collision occurs, in which case neither packet may be decoded successfully.

Performance of random access techniques is typically characterized by the *throughput* of the system. The throughput, which is unit less, is defined as the ratio of the average number of packets successfully transmitted in any given time interval divided by the number of attempted transmissions in that interval. The throughput thus equals the offered load multiplied by the probability of successful packet reception (successful packet reception), where this probability is a function of the random access protocol in use as well as the channel characteristics, which can cause packet errors in the absence of collisions. The goal of a random access method is to make the throughput as large as possible in order to fully utilize the underlying link rates.

16.14 Packet radio protocols

ALOHA, slotted ALOHA, and carrier sense multiple access (CSMA) protocols are the packet radio protocols. Random access techniques were pioneered by Abramson with the ALOHA protocol, where data is packetized and users send packets whenever they have data to send. ALOHA is very inefficient due to collisions between users, which lead to very low throughput. The throughput

can be doubled by slotting time and synchronizing the users, but even then, collisions lead to relatively low-throughput values. Modifications of ALOHA protocols are to avoid collisions and thereby increase the throughput include carrier sensing, collision detection, and collision avoidance. Long bursts of packets can be scheduled to avoid collisions, but this typically takes additional overhead. In this section, we will describe the various techniques for random access, their performance, and their comparison.

16.14.1 Pure ALOHA

Pure or unslotted ALOHA is the simplest protocol in which the user transmits data packets whenever it has data.

Whenever two frames from different users try to occupy the channel at the same time, there will be a collision and colliding frames will be destroyed. If the first bit of a new frame overlaps with just the last bit of a frame almost finished, both frames will be totally destroyed and both will have to be retransmitted later. This is shown in the Figure 16.33. Here all frames assumed to be same length because the throughput of ALOHA systems is maximized by having a uniform frame size rather than by allowing variable length frames. A frame will not suffer a collision if no other frames are sent within one frame time of its start.

Let t be the time required to send a frame. If any other user has generated a frame (Frame 1) between time t_0 and $t_0 + t$, the end of that frame will collide with the beginning of the already transmitted frame (Frame 3). Similarly, any other frame (Frame 2) started between $t_0 + t$ and $t_0 + 2t$ will bump into the end of the transmitted frame (Frame 3). The vulnerable period which is defined as the time interval during which the packets are susceptible to collisions with transmissions from other users. Each transmitted frame has a vulnerable period equal to two frame times. There will be no collision if and only if no other frame is transmitted within the vulnerable period (Figure 16.34).

The normalized throughput S of the pure ALOHA protocol in terms of the offered load is given by

$$S = Ge^{-2G} \quad (16.20)$$

Traffic load

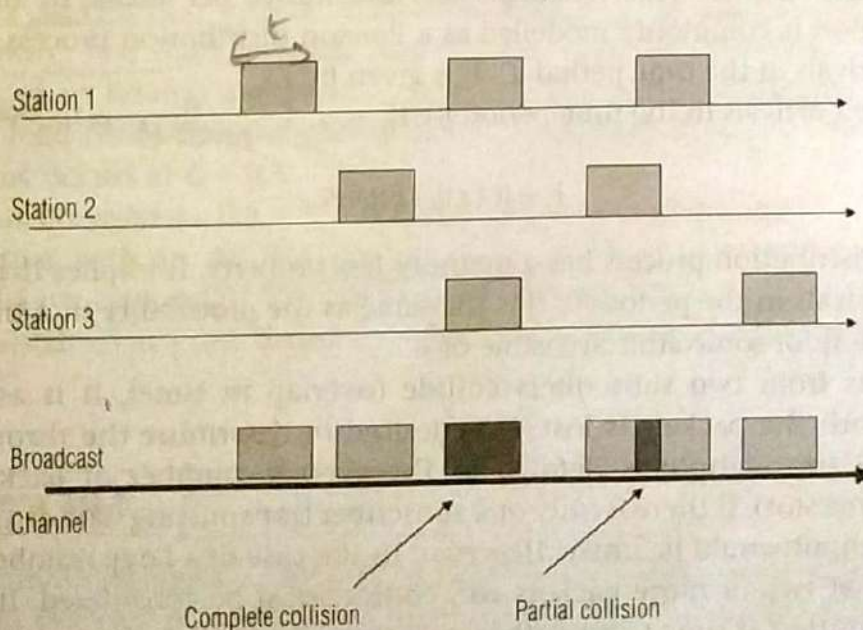


Figure 16.33 Collision in pure ALOHA

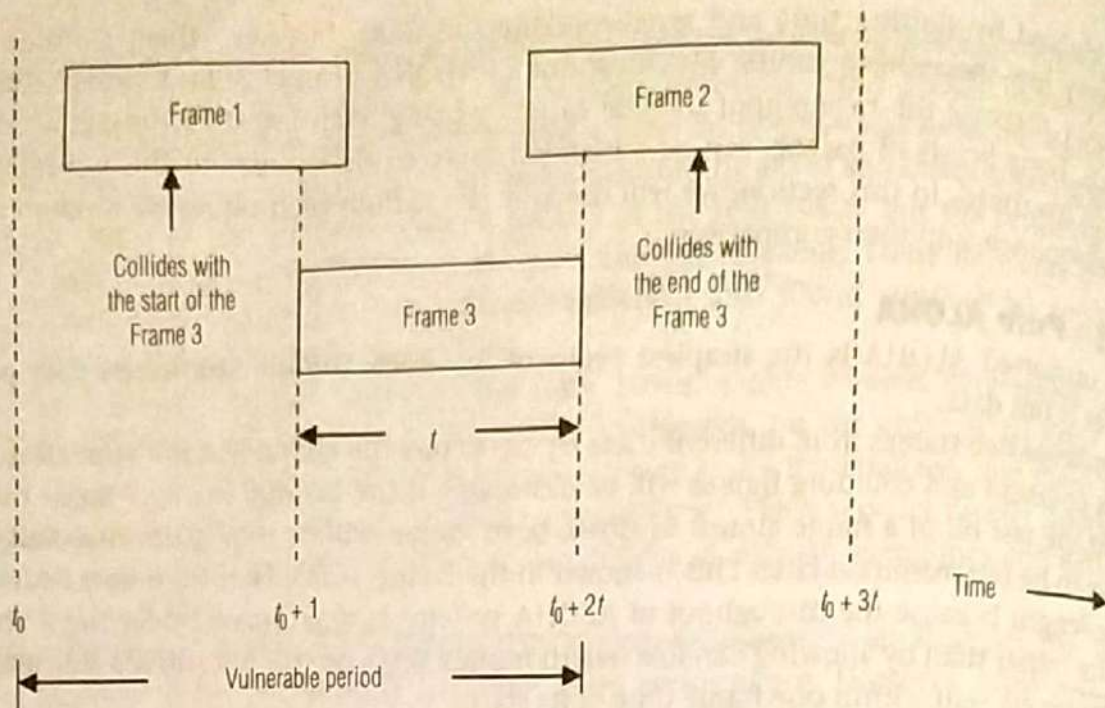


Figure 16.34 Vulnerable periods in pure ALOHA

From Equation (16.20), it should be noted that the maximum throughput occurs at traffic load $G = 50$ per cent and is $S = 1/2e$. This is about 0.184. Thus, the best channel utilization with the pure ALOHA protocol is only 18.4 per cent. At smaller offered load, channel capacity is under used and at higher offered load, too many collisions occur reducing the throughput.

Throughput of pure ALOHA

① To evaluate the performance of pure ALOHA protocol technique in terms of throughput, it is assumed that all data packets transmitted by all subscribers in the system have a constant and a fixed packet length, L . Let the average transmission rate or mean arrival rate be λ (packets/s). This means that there are, on an average, λ data packets transmitted per second by all the subscribers. This type of situation is commonly modelled as a Poisson distribution process. The probability that there are k arrivals in the time period $[0, t]$ is given by

Probability for k arrivals in the time period $[0, t]$,

$$P_k = \frac{(\lambda t)^k}{k!} e^{-\lambda t} \quad (16.21)$$

The Poisson distribution process has a memory less property. It implies that the probability that there are k arrivals in the period $[0, t]$ is the same as the probability that there are k arrivals in the period $[s, t + s]$ for some arbitrary value of s .

If two packets from two subscribers collide (overlap in time), it is assumed that the information in both the packets is lost. It is desired to determine the throughput S of this common channel (throughput is defined as the average number of packets that can be transmitted per time slot). If there is only one subscriber transmitting data at any time then the maximum throughput would be unity. However, in the case of a large number of subscribers, the probability that two or more packets will collide must be considered. It is obvious that for a packet transmitted at time t_0 , any other packet transmitted in the interval $[t_0 - T, t_0 + T]$,

where T is the packet duration, will cause a collision. With a Poisson distribution process for packet arrival times, if a packet is being transmitted, then the probability that no additional packets arrive during the period $[t_0 - T, t_0 + T]$, or $[t = 2T]$ is given by substituting $k = 0$ and $t = 2T$ in Equation (16.20). Probability that a packet is successfully received in time $2T$, $P_s = [(\lambda t) 0 / 0!] e^{-2\lambda T}$

Or, probability that a packet is successfully received in time $2T$,

$$P_s = e^{-2\lambda T} \quad (16.22)$$

Consequently, the throughput S of an ALOHA system is given by the product of the packet arrival rate λ and the probability that a packet is successfully received in time $2T$, P_s , that is,

Throughput of an ALOHA system

$$S(\text{ALOHA}) = \lambda e^{-2\lambda T} \quad (16.23)$$

Normalizing this equation to packets received per packet time T , the normalized throughput, $S(\text{ALOHA})$ is given as Normalized throughput of an ALOHA system,

$$S(\text{ALOHA}) = \lambda T e^{-2\lambda T} \quad (16.24)$$

or,

$$S(\text{ALOHA}) = G e^{-2G} \quad (16.25)$$

normalized channel traffic

where $G = \lambda T$ is the traffic occupancy or the normalized channel traffic (measured in Erlangs)

The parameter G is defined as the normalized loading per packet period, that is, the average number of packets per slot time T , due to arriving and buffered packets. It is a relative measure of the channel utilization. If $G > 1$ then the packets generated by the subscribers exceed the maximum transmission rate of the channel. It implies that in order to obtain a reasonable acceptable throughput, the rate at which new packets are generated must lie within $0 < G < 1$. Under normal loading conditions, the throughput T is the same as the total offered load, L . The total offered load L is the sum of the newly generated packets and the retransmitted packets that suffered collisions in previous transmissions. The normalized throughput is always less than or equal to unity and may be considered as the fraction of time (or fraction of an Erlang) a channel is utilized. The normalized throughput is given as the total offered load times the probability of successful transmission. The peak normalized throughput S_0 peak occurs at $G = 0.5$.

Therefore, peak throughput $= 0.5 e^{-2(0.5)} = 0.184$ packets per packet time

This means that with an ALOHA random-access channel, the maximum throughput is approximately 18.4 per cent or less than 19 per cent of the full channel capacity. In practice, the throughput is maintained at a much smaller value so as to ensure stability of the random-access approach.

Example problem 16.8

Calculate the maximum throughput of a pure ALOHA network with a large number of subscribers and a transmission rate of 2 Mbps. What is the throughput of the ALOHA network if only one subscriber is active?

Solution

For a large number of subscribers, each using a transmission rate of 1 Mbps to access a base station using ALOHA protocol, the maximum data rate that successfully passes through to the base station is given by

Maximum throughput of a pure ALOHA system, $S_{\max}(\text{ALOHA})$,

$$S_{\max}(\text{ALOHA}) = (G e^{-2G}) \times \text{transmission rate, for } G = 0.5$$

$$\begin{aligned} \text{Therefore, } S_{\max}(\text{ALOHA}) &= (0.5 e^{-2(0.5)}) \times 2 \text{ Mbps} \\ &= 0.184 \times 2 \text{ Mbps} \\ &= \mathbf{368 \text{ Kbps}} \end{aligned}$$

If only one subscriber is active in the ALOHA network, there cannot be any collision and the only subscriber can transmit all the time. Therefore,

$$\text{Throughput of the ALOHA network at 2 Mbps rate} = \mathbf{2 \text{ Mbps}}$$

16.14.2 Slotted ALOHA

In 1972, Roberts published a method for doubling the capacity of an ALOHA system. In the slotted-ALOHA system, the transmission time is divided into time slot, each slot corresponding to one frame transmission time. Users are synchronized to the time slots and the user can transmit the frame only at the beginning of a slot. This reduces the vulnerable period from $2t$ to t , that is, one frame duration and improves efficiency by reducing the probability of collision as shown in Figure 16.35.

The normalized throughput S of the slotted-ALOHA protocol in terms of the offered load is given by

$$S = G e^{-G} \quad (16.26)$$

The maximum throughput occurs at traffic load $G = 1$ and equal to $1/e$. This is about 0.368. Thus, the best channel utilization with the slotted ALOHA protocol is only 36.8 per cent, which is twice than that of pure ALOHA protocol.

Throughput of slotted ALOHA

With slotted ALOHA, a collision occurs only if the two subscribers transmit during the same time slot T . Since the sequence of the newly generated and retransmitted packets follows Poisson distribution process, the probability of successful transmission is given by

$$P_s S = e^{-\lambda T} \quad (16.27)$$

Consequently, the throughput SS of a slotted ALOHA system is given by the product of the packet arrival rate and the probability that a packet is successfully received; that is, throughput of a slotted ALOHA system,

$$S = \lambda e^{-\lambda T} \quad (16.28)$$

Normalizing this equation to packets received per packet time T , and then the normalized throughput is given by normalized throughput of a slotted ALOHA system,

$$S = \lambda T e^{-\lambda T} \quad (16.29)$$

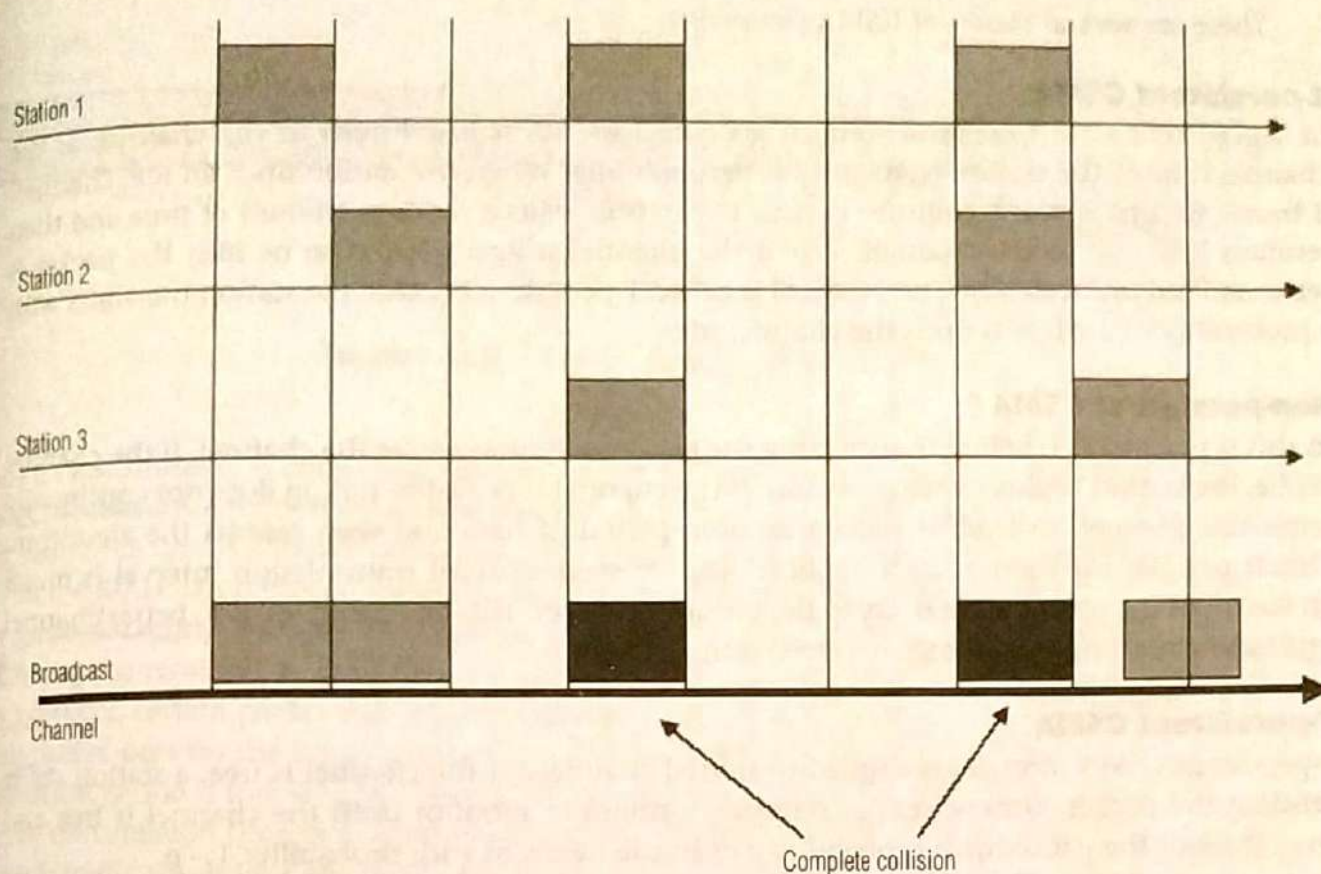


Figure 16.35 Probability of collision in slotted ALOHA

Using the definition of the normalized offered load $G = \lambda T$,

$$S(\text{Slotted ALOHA}) = Ge^{-G} \quad (16.30)$$

The probability that no other data packets will be generated during the vulnerable period is e^{-G} . The peak throughput with slotted ALOHA occurs at $G = 1$. Therefore, $S(\text{Slotted ALOHA}) = 1/e$. The peak throughput with slotted ALOHA occurs at $G = 1$. Therefore, $S(\text{Slotted ALOHA}) = 1/e \approx 0.368$ packets per packet time. That is, with a slotted ALOHA random-access protocol, the maximum throughput is 36.8 per cent of the full channel capacity. This is double that of pure or unslotted ALOHA. In other words, the slotted ALOHA protocol provides a maximum channel utilization of 0.368 Erlangs, double that of pure ALOHA protocol. Thus, the maximum throughputs of pure ALOHA and slotted ALOHA protocols are equal to 0.184 and 0.368, respectively.

16.14.3 Carrier sense multiple access protocols

The name carrier sense terms from the fact that the existence of the carrier wave on the communication channel is sensed by the user. In CSMA, a user having data to transmit first listens to the channel to check whether another transmission is in progress or not. The user starts sending only when the channel is idle, that is, there is no carrier.

There are several *classes* of CSMA protocols:

1-persistent CSMA

In 1-persistent CSMA, when a station has data to send, it first listens to the channel. If the channel is busy, the station waits until it becomes idle. When the station finds an idle channel, it transmits a packet. If a collision occurs, the station waits a random amount of time and then resumes listening to the channel. When the channel is again found to be idle, the packet is retransmitted immediately. The protocol is called 1-persistent because the station transmits with a probability of 1 when it finds the channel idle.

Non-persistent CSMA

In this type of CSMA, before transmitting the packet, a station senses the channel. If the channel is idle, the station begins its transmission. If the channel is busy, the station does not continually sense the channel. Instead, it waits a random period of time and then repeats the algorithm. This is popular for wireless LAN applications, where the packet transmission interval is much greater than the propagation delay to the farthest user. This algorithm leads to better channel utilization but longer delays than 1-persistent CSMA.

P-persistent CSMA

P-persistent CSMA protocol is applied to slotted channels. If the channel is free, a station starts sending the packet. Otherwise, the station continues to monitor until the channel is free and then it sends the packet with probability p or in the next slot with probability $1 - p$.

CSMA/CD

CSMA with collision detection (CSMA/CD) is the modification of pure CSMA protocol. This protocol is used to improve the performance of CSMA by terminating transmission as soon as a collision is detected, thus reducing the probability of a second collision on retry. As the collided frames are terminated quickly, this protocol saves time and bandwidth. This is widely used on LANs in the medium access control (MAC) sub-layer.

16.14.4 Reservation protocols

With the pure ALOHA and slotted ALOHA protocols, users monitor the broadcast channel for the success or failure of their own transmission, which give rise to a maximum channel throughput of $1/(2e)$ and $1/e$ respectively. For broadcast networks with a very short propagation delay (relative to a packet transmission time), CSMA protocols were devised to significantly improve channel throughput by requiring the users to do carrier sensing before attempting a transmission.

However, the relatively large propagation delay (0.27 s) of a satellite channel causes carrier sensing impractical except in the case of an extremely long packet or low speed channel. The following reservation protocols were proposed to improve the throughput of a satellite channel beyond that of slotted ALOHA. Although it was invented for a satellite channel, the reservation protocols can be used for any of the other broadcast media.

Reservation ALOHA

Reservation ALOHA packet access scheme shown in Figure 16.36, is an improvement over slotted ALOHA and is a combination of slotted ALOHA and TDM. In this protocol, it is possible to reserve slots for the transmission of packets and the reservation can be permanent or can be reserved on request. In one type of reservation ALOHA, the user can reserve a slot permanently.

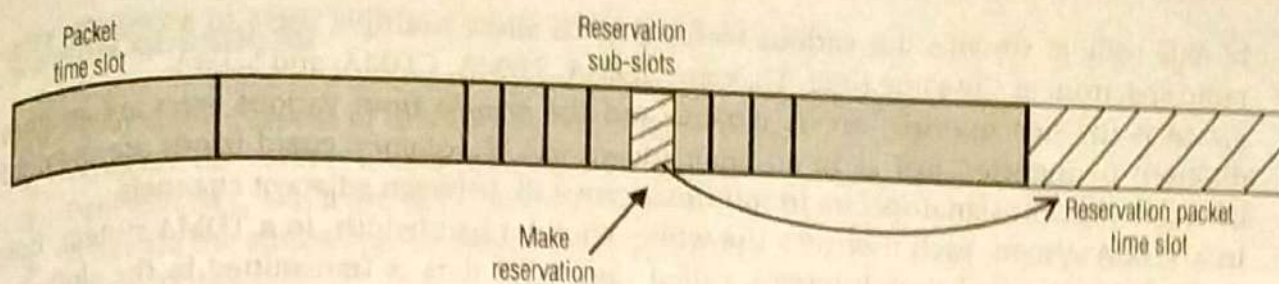


Figure 16.36 Protocol of reservation ALOHA

until its transmission is completed successfully. In another type of reservation ALOHA, it allows a user to transmit a request on a sub-slot which is reserved in each frame. If the transmission is successful, the user is allocated the next regular slot in the frame for data transmission.

In the R-ALOHA protocol, time slots are divided into contention periods as well as contention-free periods. During the contention period, a subscriber uses very short packets to contend for the upcoming contention-free intervals that will be used for transmission of longer data packets. In this protocol, certain packet slots are assigned with priority. It is possible for subscribers to reserve some packet slots for the transmission of data packets either permanently or on request basis. The reservation period is followed by a transmission period. Usually, a very small data packet is sent to reserve transmission time slot. Reservations of time slots on request basis offers better throughput for high traffic conditions although this scheme causes a larger delay even under a light load.

Packet reservation multiple access

Packet reservation multiple access (PRMA) is a multiple access scheme with frames of a fixed number of slots. If a user has a series of data packets (or speech segments) to transmit, it competes for access in any free slot. If it successfully acquires the base station, the terminal achieves reservation in the corresponding slots of the next frames, until it delivers the reservation.

16.15 Capture effect in packet radio

When packet radio multiple access techniques are used with FM or spread-spectrum modulation, the strongest user may successfully capture the intended receiver, even when many other users are also transmitting. In addition, the closest transmitter may be able to capture a receiver because of the small propagation path loss. The capture effect refers to the fact that a radio receiver may successfully decode a radio signal from one user despite the presence of interference from other users. The capture effect can be analysed using a parameter called capture ratio, which is defined as the ratio of the received signal power to the total interference power.

16.16 Summary

- The radio spectrum is a limited resource. Multiple access refers to techniques that enable multiple users to share the limited network resources efficiently. When there is more than one user to access such limited bandwidth, a multiple access scheme must be put in place to control the share of bandwidth among multiple users so that everyone can use services provided by the network and to make sure that no single user spends all available resources.